

Supporting information for:

Quantum electrodynamics effects in rovibrational spectra of molecular hydrogen

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Components of dissociation energy of H₂

The following tables collect components of the dissociation energy (in cm⁻¹) for all rovibrational levels of H₂. The abbreviations used within the table:

ν – vibrational quantum number

J – rotational quantum number

BO – the Born-Oppenheimer component

AD – the adiabatic correction

NA – the nonadiabatic correction

α^2 – the relativistic correction

α^3 – the leading QED correction

α^4 – the QED correction of the order α^4 and higher

Total – the total dissociation energy, D_0

δ – the uncertainty assigned to D_0

Components of dissociation energy of H₂

$v \setminus J$	0	1	2	3	4	5	6	7	8	9	10	11
0 BO	36112.5927	35994.0371	35758.0142	35406.6660	34943.1216	34371.4040	33696.3152	32923.3094	32058.3583	31107.8190	30078.3065	28976.5781
0 AD	5.7711	5.8348	5.9611	6.1481	6.3931	6.6924	7.0419	7.4367	7.8718	8.3418	8.8411	9.3642
0 NA	.4340	.4407	.4540	.4740	.5006	.5338	.5736	.6200	.6729	.7323	.7981	.8704
0 α^2	-.5318	-.5344	-.5395	-.5470	-.5568	-.5686	-.5822	-.5973	-.6138	-.6312	-.6494	-.6680
0 α^3	-.1948	-.1938	-.1918	-.1888	-.1849	-.1801	-.1745	-.1682	-.1613	-.1538	-.1458	-.1375
0 α^4	-.0016	-.0016	-.0016	-.0016	-.0015	-.0015	-.0014	-.0014	-.0013	-.0013	-.0012	-.0012
0 Total	36118.0696	35999.5827	35763.6964	35412.5507	34949.2721	34377.8800	33703.1726	32930.5991	32066.1266	31116.1067	30087.1493	28986.0061
0 δ	.0011	.0011	.0011	.0011	.0011	.0011	.0012	.0012	.0012	.0012	.0013	.0013
1 BO	31949.1892	31836.5534	31612.3337	31278.6001	30838.3755	30295.5438	29654.7385	28921.2187	28100.7394	27199.4234	26223.6397	25179.8929
1 AD	7.1740	7.2311	7.3443	7.5119	7.7310	7.9983	8.3096	8.6605	9.0458	9.4605	9.8991	10.3562
1 NA	1.2705	1.2761	1.2875	1.3045	1.3272	1.3556	1.3897	1.4295	1.4750	1.5264	1.5835	1.6465
1 α^2	-.5552	-.5576	-.5622	-.5691	-.5780	-.5877	-.6011	-.6148	-.6297	-.6453	-.6616	-.6781
1 α^3	-.1735	-.1726	-.1707	-.1679	-.1642	-.1597	-.1544	-.1485	-.1420	-.1350	-.1276	-.1198
1 α^4	-.0014	-.0014	-.0014	-.0014	-.0014	-.0013	-.0013	-.0012	-.0012	-.0011	-.0011	-.0010
1 Total	31956.9034	31844.3290	31620.2311	31286.6781	30846.6902	30304.1479	29663.6809	28930.5440	28110.4874	27209.6288	26234.3321	25191.0967
1 δ	.0017	.0017	.0017	.0017	.0017	.0018	.0018	.0018	.0019	.0019	.0020	.0020
2 BO	28021.4345	27914.5889	27701.9153	27385.4158	26968.0130	26453.4605	25846.2343	25151.4124	24374.5496	23521.5539	22598.5709	21611.8782
2 AD	8.3336	8.3840	8.4839	8.6314	8.8240	9.0583	9.3304	9.6358	9.9697	10.3271	10.7026	11.0910
2 NA	2.0270	2.0318	2.0413	2.0557	2.0748	2.0989	2.1278	2.1618	2.2007	2.2449	2.2942	2.3488
2 α^2	-.5731	-.5753	-.5795	-.5857	-.5937	-.6034	-.6145	-.6268	-.6400	-.6539	-.6682	-.6826
2 α^3	-.1537	-.1528	-.1510	-.1484	-.1449	-.1407	-.1358	-.1302	-.1242	-.1176	-.1106	-.1033
2 α^4	-.0013	-.0013	-.0013	-.0012	-.0012	-.0012	-.0011	-.0011	-.0011	-.0010	-.0010	-.0009
2 Total	28031.0670	27924.2753	27711.7087	27395.3676	26978.1720	26463.8724	25856.9411	25162.4519	24385.9548	23533.3534	22610.7880	21624.5312
2 δ	.0024	.0024	.0024	.0024	.0024	.0025	.0025	.0025	.0026	.0026	.0026	.0027
3 BO	24324.4499	24223.3066	24022.0055	23722.4851	23327.5744	22840.9061	22266.8106	21610.1998	20876.4463	20071.2651	19200.6029	18270.5391
3 AD	9.2420	9.2855	9.3714	9.4982	9.6632	9.8633	10.0945	10.3525	10.6327	10.9300	11.2393	11.5554
3 NA	2.7087	2.7127	2.7206	2.7325	2.7485	2.7686	2.7928	2.8214	2.8543	2.8917	2.9337	2.9804
3 α^2	-.5854	-.5874	-.5911	-.5967	-.6038	-.6124	-.6222	-.6330	-.6445	-.6565	-.6688	-.6811
3 α^3	-.1353	-.1344	-.1327	-.1303	-.1270	-.1231	-.1185	-.1133	-.1076	-.1014	-.0949	-.0881
3 α^4	-.0011	-.0011	-.0011	-.0011	-.0011	-.0010	-.0010	-.0010	-.0009	-.0009	-.0008	-.0008
3 Total	24335.6787	24234.5818	24033.3726	23733.9878	23339.2543	22852.8015	22278.9563	21622.6265	20889.1803	20084.3279	19214.0114	18284.3049
3 δ	.0031	.0031	.0031	.0031	.0031	.0031	.0032	.0032	.0032	.0033	.0033	.0034
4 BO	20855.2072	20759.7246	20569.7150	20287.0583	19914.4976	19455.5548	18914.4286	18295.8815	17605.1247	16847.7045	16029.3982	15156.1198
4 AD	9.8890	9.9251	9.9966	10.1016	10.2378	10.4019	10.5902	10.7985	11.0221	11.2561	11.4954	11.7349
4 NA	3.3187	3.3219	3.3283	3.3379	3.3508	3.3670	3.3867	3.4100	3.4369	3.4677	3.5023	3.5408
4 α^2	-.5918	-.5935	-.5968	-.6017	-.6079	-.6153	-.6238	-.6330	-.6428	-.6528	-.6629	-.6729
4 α^3	-.1181	-.1173	-.1157	-.1134	-.1104	-.1067	-.1024	-.0975	-.0922	-.0865	-.0804	-.0741
4 α^4	-.0010	-.0010	-.0010	-.0010	-.0010	-.0009	-.0009	-.0009	-.0008	-.0008	-.0007	-.0007
4 Total	20867.7039	20772.2598	20582.3263	20299.7818	19927.3669	19468.6008	18927.6785	18309.3586	17618.8479	16861.6882	16043.6518	15170.6479
4 δ	.0037	.0037	.0037	.0037	.0037	.0038	.0038	.0038	.0038	.0039	.0039	.0039
5 BO	17612.7145	17522.9036	17344.2107	17078.4626	16728.3253	16297.2225	15789.2368	15209.0018	14561.5917	13852.4141	13087.1119	12271.4764
5 AD	10.2629	10.2915	10.3476	10.4298	10.5356	10.6619	10.8049	10.9606	11.1244	11.2915	11.4566	11.6146
5 NA	3.8575	3.8599	3.8647	3.8719	3.8815	3.8938	3.9086	3.9262	3.9466	3.9699	3.9962	4.0254
5 α^2	-.5920	-.5934	-.5962	-.6003	-.6055	-.6118	-.6188	-.6263	-.6342	-.6422	-.6500	-.6573
5 α^3	-.1021	-.1014	-.0999	-.0977	-.0949	-.0915	-.0874	-.0829	-.0780	-.0727	-.0670	-.0612
5 α^4	-.0009	-.0009	-.0009	-.0009	-.0008	-.0008	-.0008	-.0007	-.0007	-.0007	-.0006	-.0006
5 Total	17626.1400	17536.3593	17357.7260	17092.0653	16742.0411	16311.0741	15803.2433	15223.1787	14575.9499	13866.9601	13101.8471	12286.3973
5 δ	.0043	.0043	.0043	.0043	.0043	.0043	.0043	.0043	.0044	.0044	.0044	.0045

Components of dissociation energy of H₂, cont.

$v \setminus J$	0	1	2	3	4	5	6	7	8	9	10	11
6 BO	14598.2891	14514.2237	14346.9982	14098.3935	13771.0098	13368.1874	12893.9129	12352.7160	11749.5653	11089.7678	10378.8782	9622.6208
6 AD	10.3525	10.3729	10.4130	10.4712	10.5449	10.6313	10.7268	10.8271	10.9279	11.0242	11.1110	11.1829
6 NA	4.3218	4.3233	4.3262	4.3306	4.3366	4.3441	4.3532	4.3639	4.3763	4.3903	4.4058	4.4227
6 α^2	-5852	-5864	-5887	-5920	-5962	-6011	-6065	-6123	-6181	-6237	-6289	-6333
6 α^3	-0873	-0866	-0852	-0832	-0806	-0774	-0736	-0694	-0649	-0599	-0547	-0493
6 α^4	-0008	-0008	-0008	-0007	-0007	-0007	-0007	-0006	-0006	-0006	-0005	-0005
6 Total	14612.2901	14528.2461	14361.0627	14112.5194	13785.2138	13382.4836	12908.3120	12367.2247	11764.1860	11104.4981	10393.7108	9637.5433
6 δ	.0048	.0048	.0048	.0048	.0048	.0048	.0048	.0048	.0048	.0048	.0049	.0049
7 BO	11815.9479	11737.7779	11582.3242	11351.3310	11047.3466	10673.6480	10234.1517	9733.3170	9176.0486	8567.6048	7913.5170	7219.5226
7 AD	10.1486	10.1606	10.1839	10.2169	10.2572	10.3020	10.3476	10.3901	10.4250	10.4474	10.4524	10.4343
7 NA	4.7029	4.7032	4.7037	4.7045	4.7055	4.7068	4.7082	4.7096	4.7109	4.7118	4.7120	4.7108
7 α^2	-5709	-5717	-5734	-5759	-5789	-5824	-5860	-5897	-5932	-5962	-5984	-5995
7 α^3	-0736	-0729	-0716	-0698	-0673	-0643	-0609	-0570	-0528	-0483	-0435	-0386
7 α^4	-0007	-0007	-0006	-0006	-0006	-0006	-0006	-0005	-0005	-0005	-0004	-0004
7 Total	11830.1543	11751.9964	11596.5661	11365.6062	11061.6626	10688.0095	10248.5600	9747.7694	9190.5379	8582.1192	7928.0390	7234.0292
7 δ	.0052	.0052	.0052	.0052	.0052	.0052	.0052	.0052	.0052	.0052	.0052	.0052
8 BO	9272.9561	9200.9278	9057.7445	8845.1263	8565.5890	8222.3732	7819.3619	7360.9906	6852.1602	6298.1569	5704.5851	5077.3199
8 AD	9.6483	9.6517	9.6577	9.6648	9.6707	9.6726	9.6670	9.6500	9.6172	9.5638	9.4847	9.3741
8 NA	4.9838	4.9824	4.9796	4.9752	4.9692	4.9614	4.9516	4.9394	4.9242	4.9054	4.8821	4.8528
8 α^2	-5479	-5485	-5495	-5509	-5527	-5545	-5562	-5575	-5583	-5582	-5569	-5540
8 α^3	-0609	-0603	-0591	-0574	-0551	-0524	-0492	-0456	-0418	-0377	-0333	-0289
8 α^4	-0006	-0006	-0005	-0005	-0005	-0005	-0005	-0004	-0004	-0004	-0003	-0003
8 Total	9286.9790	9214.9526	9071.7726	8859.1575	8579.6206	8236.3999	7833.3746	7374.9763	6866.1012	6312.0299	5718.3612	5090.9637
8 δ	.0055	.0055	.0055	.0055	.0055	.0054	.0054	.0054	.0054	.0054	.0054	.0053
9 BO	6980.5984	6915.0837	6784.9238	6591.8323	6338.3203	6027.6323	5663.6708	5250.9187	4794.3657	4299.4465	3771.9991	3218.2507
9 AD	8.8590	8.8537	8.8422	8.8232	8.7945	8.7533	8.6963	8.6196	8.5188	8.3891	8.2250	8.0203
9 NA	5.1365	5.1326	5.1246	5.1125	5.0959	5.0744	5.0473	5.0139	4.9730	4.9232	4.8624	4.7879
9 α^2	-5151	-5153	-5155	-5159	-5161	-5159	-5153	-5139	-5114	-5075	-5019	-4940
9 α^3	-0491	-0486	-0475	-0459	-0439	-0414	-0385	-0353	-0318	-0281	-0242	-0202
9 α^4	-0005	-0005	-0004	-0004	-0004	-0004	-0004	-0003	-0003	-0003	-0003	-0002
9 Total	6994.0292	6928.5057	6798.3271	6605.2057	6351.6504	6040.9022	5676.8602	5264.0027	4807.3140	4312.2229	3784.5601	3230.5444
9 δ	.0056	.0056	.0056	.0056	.0056	.0056	.0055	.0055	.0054	.0054	.0053	.0052
10 BO	4955.2699	4896.8118	4780.7739	4608.8912	4383.7137	4108.5514	3787.4126	3424.9438	3026.3806	2597.5217	2144.7391	1675.0460
10 AD	7.8021	7.7882	7.7597	7.7151	7.6525	7.5691	7.4616	7.3260	7.1577	6.9514	6.7003	6.3963
10 NA	5.1160	5.1082	5.0925	5.0684	5.0353	4.9925	4.9385	4.8719	4.7903	4.6909	4.5694	4.4200
10 α^2	-4706	-4704	-4698	-4687	-4671	-4646	-4612	-4563	-4498	-4411	-4297	-4151
10 α^3	-0384	-0379	-0369	-0355	-0336	-0313	-0288	-0259	-0228	-0195	-0161	-0126
10 α^4	-0004	-0004	-0004	-0003	-0003	-0003	-0003	-0003	-0002	-0002	-0002	-0002
10 Total	4967.6786	4909.1996	4793.1191	4621.1702	4395.9005	4120.6166	3799.3225	3436.6592	3037.8559	2608.7032	2155.5628	1685.4344
10 δ	.0056	.0056	.0056	.0055	.0055	.0055	.0054	.0053	.0052	.0051	.0050	.0048
11 BO	3220.0418	3169.4253	3069.1017	2920.8723	2727.3991	2492.1651	2219.4373	1914.2414	1582.3694	1230.4417	866.0734	498.2452
11 AD	6.5140	6.4918	6.4469	6.3777	6.2821	6.1573	5.9996	5.8045	5.5658	5.2756	4.9227	4.4892
11 NA	4.8565	4.8428	4.8151	4.7725	4.7141	4.6380	4.5419	4.4225	4.2753	4.0940	3.8690	3.5852
11 α^2	-4122	-4114	-4097	-4069	-4030	-3976	-3905	-3812	-3692	-3538	-3342	-3088
11 α^3	-0285	-0281	-0272	-0259	-0243	-0223	-0200	-0175	-0148	-0119	-0091	-0062
11 α^4	-0003	-0003	-0003	-0003	-0003	-0002	-0002	-0002	-0002	-0001	-0001	-0001
11 Total	3230.9712	3180.3202	3079.9265	2931.5894	2737.9677	2502.5403	2229.5680	1924.0695	1591.8264	1239.4455	874.5218	506.0045
11 δ	.0053	.0053	.0053	.0052	.0052	.0051	.0050	.0048	.0047	.0045	.0042	.0039

Components of dissociation energy of H₂, cont.

$v \setminus J$	0	1	2	3	4	5	6	7	8	9	10
12 BO	1806.9489	1765.3260	1683.0605	1562.1049	1405.3732	1216.7368	1001.0464	764.2101	513.3908	257.4738	8.3253
12 AD	5.0372	5.0068	4.9451	4.8505	4.7203	4.5506	4.3360	4.0685	3.7358	3.3166	2.7633
12 NA	4.2658	4.2429	4.1966	4.1254	4.0271	3.8984	3.7345	3.5281	3.2682	2.9355	2.4879
12 α^2	-.3365	-.3350	-.3319	-.3269	-.3199	-.3106	-.2983	-.2825	-.2621	-.2355	-.1992
12 α^3	-.0196	-.0192	-.0184	-.0173	-.0159	-.0142	-.0122	-.0101	-.0078	-.0055	-.0031
12 α^4	-.0002	-.0002	-.0002	-.0002	-.0002	-.0002	-.0001	-.0001	-.0001	-.0001	-.0001
12 Total	1815.8955	1774.2213	1691.8517	1570.7364	1413.7847	1224.8610	1008.8062	771.5140	520.1248	263.4848	13.3741
12 δ	.0047	.0046	.0046	.0045	.0044	.0043	.0041	.0039	.0036	.0032	.0027
13 BO	760.3903	729.5279	668.9437	580.9383	469.0254	338.0600	194.5461	47.4825			
13 AD	3.3933	3.3526	3.2697	3.1417	2.9631	2.7253	2.4123	1.9886			
13 NA	3.2222	3.1851	3.1096	2.9928	2.8295	2.6114	2.3232	1.9317			
13 α^2	-.2390	-.2365	-.2313	-.2232	-.2116	-.1960	-.1750	-.1460			
13 α^3	-.0115	-.0112	-.0106	-.0096	-.0084	-.0070	-.0053	-.0036			
13 α^4	-.0001	-.0001	-.0001	-.0001	-.0001	-.0001	-.0001	.0000			
13 Total	766.7551	735.8177	675.0810	586.8399	474.5979	343.1936	199.1012	51.2532			
13 δ	.0035	.0035	.0034	.0033	.0031	.0029	.0025	.0021			
14 BO	141.7951	124.7523	92.3077	48.0033	— ^a						
14 AD	1.5343	1.4739	1.3479	1.1416	— ^a						
14 NA	1.5847	1.5226	1.3934	1.1825	.0888						
14 α^2	-.1132	-.1090	-.1000	-.0854	-.0606						
14 α^3	-.0044	-.0041	-.0036	-.0028	-.0016						
14 α^4	-.0001	.0000	.0000	.0000	.0000						
14 Total	144.7964	127.6357	94.9453	50.2393	.0265						
14 δ	.0017	.0017	.0015	.0013	.0001						

^a Does not exist in this approximation.

Components of dissociation energy of H₂, cont.

$v \backslash J$	12	13	14	15	16	17	18	19	20	21	22	23
0 BO	27809.4302	26583.6105	25305.7462	23982.2860	22619.4577	21223.2371	19799.3296	18353.1603	16889.8728	15414.3355	13931.1527	12444.6818
0 AD	9.9055	10.4596	11.0212	11.5852	12.1465	12.7005	13.2425	13.7681	14.2729	14.7525	15.2029	15.6196
0 NA	.9492	1.0344	1.1260	1.2241	1.3286	1.4397	1.5573	1.6814	1.8122	1.9497	2.0940	2.2451
0 α^2	-.6868	-.7055	-.7240	-.7421	-.7595	-.7760	-.7917	-.8062	-.8196	-.8316	-.8422	-.8513
0 α^3	-.1288	-.1200	-.1109	-.1018	-.0926	-.0835	-.0743	-.0653	-.0565	-.0478	-.0393	-.0310
0 α^4	-.0011	-.0010	-.0010	-.0009	-.0008	-.0008	-.0007	-.0006	-.0005	-.0005	-.0004	-.0004
0 Total	27819.4682	26594.2780	25317.0575	23994.2505	22632.0799	21236.5171	19813.2627	18367.7376	16905.0813	15430.1579	13947.5677	12461.6638
0 δ	.0014	.0015	.0015	.0016	.0017	.0018	.0019	.0021	.0022	.0023	.0025	.0026
1 BO	24074.7260	22914.6378	21706.0160	20455.0847	19167.8656	17850.1517	16507.4922	15145.1868	13768.2885	12381.6135	10989.7585	9597.1235
1 AD	10.8264	11.3045	11.7850	12.2629	12.7333	13.1911	13.6317	14.0502	14.4420	14.8023	15.1262	15.4086
1 NA	1.7155	1.7904	1.8713	1.9584	2.0516	2.1511	2.2570	2.3693	2.4880	2.6134	2.7453	2.8839
1 α^2	-.6948	-.7112	-.7273	-.7429	-.7577	-.7715	-.7843	-.7959	-.8062	-.8150	-.8222	-.8277
1 α^3	-.1117	-.1034	-.0949	-.0864	-.0779	-.0693	-.0609	-.0525	-.0443	-.0363	-.0284	-.0208
1 α^4	-.0010	-.0009	-.0008	-.0008	-.0007	-.0006	-.0006	-.0005	-.0005	-.0004	-.0003	-.0003
1 Total	24086.4605	22926.9171	21718.8492	20468.4759	19181.8142	17864.6524	16522.5351	15160.7573	13784.3676	12398.1776	11006.7791	9614.5672
1 δ	.0021	.0021	.0022	.0023	.0024	.0025	.0026	.0027	.0029	.0030	.0031	.0033
2 BO	20567.7938	19472.5994	18332.4783	17153.4682	15941.4274	14702.0131	13440.6715	12162.6372	10872.9422	9576.4337	8277.7999	6981.6067
2 AD	11.4869	11.8851	12.2802	12.6671	13.0406	13.3957	13.7272	14.0300	14.2988	14.5281	14.7122	14.8446
2 NA	2.4089	2.4745	2.5457	2.6226	2.7053	2.7939	2.8884	2.9890	3.0955	3.2080	3.3263	3.4502
2 α^2	-.6970	-.7111	-.7247	-.7376	-.7496	-.7606	-.7703	-.7787	-.7855	-.7906	-.7939	-.7951
2 α^3	-.0958	-.0880	-.0802	-.0722	-.0643	-.0564	-.0485	-.0408	-.0332	-.0259	-.0187	-.0117
2 α^4	-.0008	-.0008	-.0007	-.0007	-.0006	-.0005	-.0005	-.0004	-.0004	-.0003	-.0002	-.0002
2 Total	20580.8960	19486.1590	18346.4985	17167.9474	15956.3587	14717.3851	13456.4677	12178.8362	10889.5174	9593.3530	8295.0256	6999.0945
2 δ	.0028	.0028	.0029	.0030	.0031	.0032	.0033	.0034	.0035	.0036	.0037	.0039
3 BO	17287.1990	16256.6838	15185.0135	14078.0862	12941.6497	11781.2863	10602.4104	9410.2757	8209.9952	7006.5729	5804.9493	4610.0657
3 AD	11.8729	12.1865	12.4908	12.7804	13.0501	13.2944	13.5077	13.6843	13.8181	13.9026	13.9306	13.8939
3 NA	3.0320	3.0885	3.1501	3.2168	3.2887	3.3658	3.4480	3.5352	3.6272	3.7237	3.8241	3.9275
3 α^2	-.6931	-.7047	-.7156	-.7257	-.7347	-.7424	-.7487	-.7534	-.7562	-.7571	-.7556	-.7515
3 α^3	-.0811	-.0739	-.0666	-.0592	-.0519	-.0446	-.0373	-.0302	-.0233	-.0166	-.0101	-.0038
3 α^4	-.0007	-.0007	-.0006	-.0006	-.0005	-.0004	-.0004	-.0003	-.0003	-.0002	-.0002	-.0001
3 Total	17301.3289	16271.1794	15199.8715	14093.2980	12957.2014	11797.1591	10618.5796	9426.7112	8226.6608	7023.4255	5821.9382	4627.1316
3 δ	.0034	.0035	.0035	.0036	.0037	.0038	.0038	.0039	.0040	.0041	.0042	.0044
4 BO	14233.8414	13268.5282	12266.0890	11232.3425	10172.9970	9093.6453	7999.7724	6896.7794	5790.0229	4684.8764	3586.8202	2501.5761
4 AD	11.9690	12.1924	12.3995	12.5849	12.7427	12.8671	12.9519	12.9905	12.9756	12.8991	12.7513	12.5208
4 NA	3.5835	3.6304	3.6814	3.7365	3.7958	3.8588	3.9254	3.9949	4.0666	4.1393	4.2113	4.2804
4 α^2	-.6824	-.6913	-.6993	-.7062	-.7118	-.7158	-.7182	-.7185	-.7166	-.7121	-.7046	-.6937
4 α^3	-.0676	-.0609	-.0541	-.0473	-.0405	-.0338	-.0272	-.0208	-.0145	-.0084	-.0026	-.0029
4 α^4	-.0006	-.0006	-.0005	-.0005	-.0004	-.0004	-.0003	-.0003	-.0002	-.0002	-.0001	-.0001
4 Total	14248.6434	13283.5982	12281.4160	11247.9099	10188.7828	9109.6212	8015.9040	6913.0252	5806.3338	4701.1940	3603.0755	2517.6864
4 δ	.0040	.0040	.0041	.0041	.0042	.0043	.0043	.0044	.0045	.0046	.0047	.0047
5 BO	11411.3755	10512.6972	9581.3093	8623.0355	7643.6478	6648.8775	5644.4446	4636.1112	3629.7676	2631.5640	1648.1214	686.8842
5 AD	11.7601	11.8873	11.9906	12.0642	12.1016	12.0965	12.0417	11.9291	11.7496	11.4918	11.1414	10.6779
5 NA	4.0576	4.0926	4.1304	4.1705	4.2125	4.2557	4.2991	4.3413	4.3801	4.4127	4.4344	4.4384
5 α^2	-.6641	-.6699	-.6746	-.6779	-.6795	-.6793	-.6768	-.6718	-.6639	-.6524	-.6367	-.6158
5 α^3	-.0551	-.0490	-.0428	-.0365	-.0303	-.0242	-.0182	-.0124	-.0068	-.0015	.0035	.0082
5 α^4	-.0005	-.0005	-.0004	-.0004	-.0003	-.0003	-.0002	-.0002	-.0001	-.0001	-.0001	.0000
5 Total	11426.4734	10527.9578	9596.7126	8638.5554	7659.2518	6664.5260	5660.0900	4651.6971	3645.2264	2646.8145	1663.0640	701.3929
5 δ	.0045	.0045	.0046	.0046	.0047	.0047	.0047	.0048	.0048	.0049	.0049	.0049

Components of dissociation energy of H₂, cont.

$v \setminus J$	12	13	14	15	16	17	18	19	20	21	22
6 BO	8826.8275	7997.3920	7140.2429	6261.3368	5366.6735	4462.3376	3554.5745	2649.9163	1755.3887	878.8631	29.7112
6 AD	11.2344	11.2597	11.2528	11.2071	11.1158	10.9712	10.7645	10.4851	10.1192	9.6476	9.0402
6 NA	4.4406	4.4593	4.4779	4.4958	4.5116	4.5235	4.5292	4.5249	4.5056	4.4629	4.3835
6 α^2	-6369	-6392	-6400	-6391	-6360	-6305	-6220	-6100	-5938	-5722	-5435
6 α^3	-0438	-0381	-0325	-0268	-0212	-0157	-0104	-0053	-0005	.0040	.0081
6 α^4	-0004	-0004	-0003	-0003	-0003	-0002	-0002	-0001	-0001	.0000	.0000
6 Total	8841.8215	8012.4332	7155.3007	6276.3735	5381.6434	4477.1860	3569.2356	2664.3109	1769.4191	892.4054	42.5995
6 δ	.0049	.0049	.0049	.0049	.0050	.0050	.0050	.0050	.0050	.0049	.0048
7 BO	6491.5172	5735.5270	4957.7045	4164.3543	3361.9951	2557.4749	1758.1694	972.3301	209.7400		
7 AD	10.3876	10.3060	10.1831	10.0114	9.7827	9.4869	9.1113	8.6376	8.0373		
7 NA	4.7076	4.7012	4.6902	4.6724	4.6450	4.6038	4.5429	4.4531	4.3193		
7 α^2	-5994	-5976	-5937	-5875	-5784	-5659	-5491	-5270	-4977		
7 α^3	-0335	-0284	-0233	-0182	-0132	-0084	-0038	.0005	.0044		
7 α^4	-0003	-0003	-0003	-0002	-0002	-0001	-0001	-0001	.0000		
7 Total	6505.9792	5749.9080	4971.9605	4178.4321	3375.8309	2570.9911	1771.2705	984.8942	221.6033		
7 δ	.0052	.0052	.0052	.0051	.0051	.0051	.0050	.0049	.0047		
8 BO	4422.4828	3746.4465	3055.8806	2357.8507	1660.0067	970.9237	300.7647				
8 AD	9.2260	9.0337	8.7892	8.4833	8.1037	7.6332	7.0443				
8 NA	4.8160	4.7692	4.7091	4.6312	4.5285	4.3910	4.2016				
8 α^2	-5493	-5424	-5326	-5194	-5020	-4793	-4493				
8 α^3	-0243	-0197	-0152	-0108	-0065	-0024	.0013				
8 α^4	-0003	-0002	-0002	-0002	-0001	-0001	-0001				
8 Total	4435.9509	3759.6871	3068.8309	2370.4348	1672.1303	982.4662	311.5626				
8 δ	.0053	.0052	.0052	.0051	.0050	.0048	.0046				
9 BO	2644.8437	2058.9184	1468.2878	881.7751	309.8952						
9 AD	7.7678	7.4587	7.0814	6.6193	6.0447						
9 NA	4.6961	4.5818	4.4374	4.2516	4.0048						
9 α^2	-4835	-4696	-4515	-4281	-3973						
9 α^3	-0162	-0122	-0083	-0046	-0011						
9 α^4	-0002	-0002	-0001	-0001	-0001						
9 Total	2656.8077	2070.4770	1479.3467	892.2132	319.5464						
9 δ	.0051	.0050	.0049	.0047	.0044						
10 BO	1196.2611	717.3517	249.1650								
10 AD	6.0280	5.5786	5.0193								
10 NA	4.2345	3.9996	3.6931								
10 α^2	-3962	-3719	-3400								
10 α^3	-0092	-0059	-0027								
10 α^4	-0001	-0001	-0001								
10 Total	1206.1180	726.5520	257.5346								
10 δ	.0046	.0044	.0040								
11 BO	138.1669										
11 AD	3.9416										
11 NA	3.2148										
11 α^2	-2753										
11 α^3	-0034										
11 α^4	-0001										
11 Total	145.0445										
11 δ	.0035										

Components of dissociation energy of H₂, concl.

$v \backslash J$	24	25	26	27	28	29	30	31
0 BO	10959.0542	9478.2014	8005.8846	6545.7303	5101.2727	3676.0048	2273.4444	897.2218
0 AD	15.9981	16.3340	16.6221	16.8572	17.0329	17.1423	17.1769	17.1261
0 NA	2.4031	2.5682	2.7404	2.9198	3.1065	3.3006	3.5019	3.7105
0 α^2	-8588	-8645	-8684	-8704	-8703	-8678	-8629	-8551
0 α^3	-0229	-0152	-0077	-0004	.0064	.0130	.0192	.0251
0 α^4	-0003	-0002	-0002	-0001	-0001	.0000	.0000	.0001
0 Total	10976.5735	9496.2237	8024.3709	6564.6364	5120.5483	3695.5928	2293.2796	917.2285
0 δ	.0028	.0030	.0031	.0033	.0035	.0037	.0039	.0041
1 BO	8207.9415	6826.3161	5456.2676	4101.7929	2766.9437	1455.9363	173.3131	
1 AD	15.6440	15.8264	15.9492	16.0045	15.9832	15.8734	15.6600	
1 NA	3.0290	3.1807	3.3389	3.5033	3.6735	3.8488	4.0284	
1 α^2	-8313	-8328	-8322	-8292	-8234	-8146	-8021	
1 α^3	-0134	-0063	.0005	.0070	.0131	.0189	.0243	
1 α^4	-0002	-0002	-0001	-0001	.0000	.0000	.0001	
1 Total	8225.7696	6844.4839	5474.7238	4120.4784	2785.7900	1474.8629	192.2238	
1 δ	.0034	.0036	.0037	.0039	.0041	.0043	.0045	
2 BO	5692.3452	4414.4956	3152.6140	1911.4560	696.1630			
2 AD	14.9184	14.9254	14.8559	14.6979	14.4356			
2 NA	3.5795	3.7136	3.8516	3.9924	4.1340			
2 α^2	-7941	-7905	-7841	-7744	-7608			
2 α^3	-0050	.0014	.0074	.0132	.0185			
2 α^4	-0001	-0001	.0000	.0000	.0000			
2 Total	5710.0438	4432.3453	3170.5449	1929.3851	713.9904			
2 δ	.0040	.0041	.0043	.0044	.0046			
3 BO	3426.9550	2260.8735	1117.5022	----- ^a				
3 AD	13.7827	13.5850	13.2850	----- ^a				
3 NA	4.0327	4.1379	4.2403	20.4754				
3 α^2	-7446	-7342	-7197	-7002				
3 α^3	.0022	.0078	.0130	.0179				
3 α^4	-0001	.0000	.0000	.0000				
3 Total	3444.0280	2277.8701	1134.3209	19.7931				
3 δ	.0045	.0046	.0047	.0223				
4 BO	1435.3165	395.0112						
4 AD	12.1920	11.7435						
4 NA	4.3428	4.3926						
4 α^2	-6786	-6582						
4 α^3	.0080	.0128						
4 α^4	.0000	.0000						
4 Total	1451.1807	410.5020						
4 δ	.0048	.0048						

^a Does not exist in this approximation.

Components of dissociation energy of D₂

The following tables collect components of the dissociation energy (in cm⁻¹) for all rovibrational levels of D₂. The abbreviations used within the table:

ν – vibrational quantum number

J – rotational quantum number

BO – the Born-Oppenheimer component

AD – the adiabatic correction

NA – the nonadiabatic correction

α^2 – the relativistic correction

α^3 – the leading QED correction

α^4 – the QED correction of the order α^4 and higher

Total – the total dissociation energy, D_0

δ – the uncertainty assigned to D_0

Components of dissociation energy of D₂

$v \setminus J$	0	1	2	3	4	5	6	7	8	9	10	11	12
0 BO	36746.1623	36686.3645	36567.0439	36388.7458	36152.2773	35858.6950	35509.2886	35105.5627	34649.2150	34142.1138	33586.2748	32983.8365	32337.0366
0 AD	2.7725	2.7886	2.8209	2.8690	2.9321	3.0110	3.1040	3.2106	3.3303	3.4621	3.6052	3.7587	3.9217
0 NA	.1563	.1580	.1614	.1666	.1734	.1819	.1921	.2040	.2176	.2329	.2498	.2684	.2887
0 α^2	-.5278	-.5291	-.5317	-.5356	-.5407	-.5469	-.5543	-.5627	-.5720	-.5822	-.5932	-.6048	-.6170
0 α^3	-.1982	-.1977	-.1967	-.1951	-.1931	-.1906	-.1877	-.1843	-.1805	-.1763	-.1718	-.1669	-.1617
0 α^4	-.0016	-.0016	-.0016	-.0016	-.0016	-.0016	-.0015	-.0015	-.0015	-.0015	-.0014	-.0014	-.0013
0 Total	36748.3634	36688.5828	36569.2963	36391.0490	36154.6478	35861.1488	35511.8412	35108.2289	34652.0088	34145.0488	33589.3635	32987.0905	32340.4669
0 δ	.0009	.0009	.0009	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0008	.0008	.0008
1 BO	33751.7214	33694.0350	33578.9306	33406.9404	33178.8522	32895.6968	32558.7327	32169.4267	31729.4340	31240.5755	30704.8145	30124.2337	29501.0119
1 AD	3.2940	3.3090	3.3389	3.3836	3.4425	3.5154	3.6015	3.7002	3.8109	3.9328	4.0648	4.2063	4.3562
1 NA	.4608	.4623	.4654	.4699	.4760	.4837	.4928	.5035	.5156	.5293	.5446	.5613	.5795
1 α^2	-.5455	-.5467	-.5492	-.5528	-.5576	-.5635	-.5704	-.5783	-.5870	-.5965	-.6068	-.6176	-.6289
1 α^3	-.1828	-.1823	-.1814	-.1799	-.1779	-.1756	-.1727	-.1695	-.1658	-.1618	-.1575	-.1529	-.1479
1 α^4	-.0015	-.0015	-.0015	-.0015	-.0015	-.0015	-.0014	-.0014	-.0014	-.0013	-.0013	-.0013	-.0012
1 Total	33754.7463	33697.0758	33582.0029	33410.0597	33182.0337	32898.9554	32562.0824	32172.8812	31733.0063	31244.2778	30708.6583	30128.2296	29505.1695
1 δ	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0008	.0008	.0008	.0008	.0008	.0008
2 BO	30876.4732	30820.8613	30709.8996	30544.1078	30324.2550	30051.3475	29726.6132	29351.4833	28927.5723	28456.6555	27940.6469	27381.5762	26781.5662
2 AD	3.7553	3.7691	3.7968	3.8379	3.8922	3.9593	4.0385	4.1292	4.2308	4.3424	4.4632	4.5924	4.7290
2 NA	.7446	.7459	.7486	.7527	.7581	.7649	.7730	.7825	.7934	.8057	.8193	.8343	.8507
2 α^2	-.5605	-.5616	-.5639	-.5673	-.5718	-.5773	-.5838	-.5911	-.5993	-.6081	-.6176	-.6277	-.6381
2 α^3	-.1682	-.1677	-.1668	-.1654	-.1635	-.1612	-.1585	-.1554	-.1520	-.1481	-.1440	-.1396	-.1348
2 α^4	-.0014	-.0014	-.0014	-.0014	-.0014	-.0013	-.0013	-.0013	-.0013	-.0012	-.0012	-.0012	-.0011
2 Total	30880.2430	30824.6456	30713.7129	30547.9643	30328.1686	30055.3317	29730.6810	29355.6472	28931.8440	28461.0461	27945.1666	27386.2346	26786.3718
2 δ	.0009	.0009	.0009	.0009	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0008	.0008
3 BO	28118.3258	28064.7586	27957.8803	27798.1984	27586.4644	27323.6608	27010.9865	26649.8387	26241.7930	25788.5817	25292.0719	24754.2435	24177.1665
3 AD	4.1553	4.1680	4.1932	4.2308	4.2804	4.3415	4.4136	4.4962	4.5884	4.6896	4.7989	4.9155	5.0384
3 NA	1.0087	1.0099	1.0123	1.0159	1.0207	1.0267	1.0339	1.0423	1.0520	1.0629	1.0751	1.0885	1.1032
3 α^2	-.5727	-.5738	-.5759	-.5791	-.5833	-.5884	-.5944	-.6012	-.6087	-.6169	-.6257	-.6349	-.6446
3 α^3	-.1543	-.1538	-.1529	-.1516	-.1498	-.1476	-.1450	-.1421	-.1388	-.1351	-.1311	-.1269	-.1224
3 α^4	-.0013	-.0013	-.0013	-.0013	-.0013	-.0012	-.0012	-.0012	-.0012	-.0011	-.0011	-.0011	-.0010
3 Total	28122.7615	28069.2076	27962.3556	27802.7132	27591.0311	27328.2917	27015.6934	26654.6328	26246.6848	25793.5810	25297.1880	24759.4845	24182.5401
3 δ	.0009	.0009	.0009	.0009	.0009	.0009	.0009	.0009	.0009	.0009	.0009	.0009	.0009
4 BO	25475.6264	25424.0816	25321.2421	25167.6042	24963.9021	24711.0959	24410.3570	24063.0502	23670.7143	23235.0410	22757.8535	22241.0848	21686.7563
4 AD	4.4925	4.5040	4.5268	4.5607	4.6054	4.6604	4.7253	4.7994	4.8821	4.9725	5.0699	5.1735	5.2824
4 NA	1.2540	1.2551	1.2572	1.2604	1.2646	1.2699	1.2762	1.2837	1.2922	1.3019	1.3126	1.3246	1.3377
4 α^2	-.5821	-.5831	-.5851	-.5880	-.5919	-.5966	-.6021	-.6084	-.6153	-.6228	-.6309	-.6393	-.6481
4 α^3	-.1411	-.1406	-.1398	-.1385	-.1368	-.1347	-.1322	-.1294	-.1262	-.1227	-.1190	-.1149	-.1106
4 α^4	-.0012	-.0012	-.0012	-.0012	-.0012	-.0011	-.0011	-.0011	-.0011	-.0010	-.0010	-.0010	-.0010
4 Total	25480.6486	25429.1157	25326.3000	25172.6976	24969.0422	24716.2938	24415.6231	24068.3944	23676.1459	23246.5687	22763.4852	22246.8277	21692.6167
4 δ	.0010	.0010	.0010	.0010	.0010	.0010	.0010	.0010	.0009	.0009	.0009	.0009	.0009
5 BO	22947.1832	22897.6464	22798.8173	22651.1814	22455.4566	22212.5818	21923.7023	21590.1524	21213.4365	20795.2084	20337.2504	19841.4522	19309.7903
5 AD	4.7652	4.7753	4.7956	4.8258	4.8654	4.9142	4.9716	5.0370	5.1098	5.1891	5.2742	5.3644	5.4586
5 NA	1.4813	1.4822	1.4840	1.4868	1.4904	1.4950	1.5005	1.5070	1.5145	1.5229	1.5324	1.5428	1.5543
5 α^2	-.5886	-.5895	-.5914	-.5941	-.5976	-.6019	-.6069	-.6127	-.6190	-.6258	-.6330	-.6406	-.6485
5 α^3	-.1285	-.1281	-.1273	-.1260	-.1244	-.1224	-.1201	-.1174	-.1143	-.1110	-.1074	-.1035	-.0995
5 α^4	-.0011	-.0011	-.0011	-.0011	-.0011	-.0010	-.0010	-.0010	-.0010	-.0010	-.0009	-.0009	-.0009
5 Total	22952.7114	22903.1852	22804.3772	22656.7728	22461.0894	22218.2657	21929.4464	21595.9654	21219.3265	20801.1827	20343.3156	19847.6143	19316.0545
5 δ	.0010	.0010	.0010	.0010	.0010	.0010	.0010	.0010	.0010	.0010	.0010	.0010	.0010

Components of dissociation energy of D₂, cont.

$v \setminus J$	0	1	2	3	4	5	6	7	8	9	10	11	12
6 BO	20532.2957	20484.7611	20389.9314	20248.2818	20060.5148	19827.5495	19550.5068	19230.6927	18869.5795	18468.7861	18030.0573	17555.2435	17046.2808
6 AD	4.9712	4.9801	4.9978	5.0240	5.0585	5.1008	5.1505	5.2068	5.2693	5.3371	5.4095	5.4856	5.5647
6 NA	1.6907	1.6915	1.6930	1.6954	1.6985	1.7024	1.7071	1.7126	1.7190	1.7262	1.7343	1.7433	1.7532
6 α^2	-.5921	-.5930	-.5946	-.5971	-.6002	-.6041	-.6087	-.6138	-.6195	-.6256	-.6320	-.6388	-.6457
6 α^3	-.1166	-.1162	-.1154	-.1142	-.1127	-.1108	-.1085	-.1059	-.1030	-.0999	-.0964	-.0928	-.0889
6 α^4	-.0010	-.0010	-.0010	-.0010	-.0010	-.0010	-.0009	-.0009	-.0009	-.0009	-.0008	-.0008	-.0008
6 Total	20538.2479	20490.7226	20395.9113	20254.2889	20066.5579	19833.6368	19556.6462	19236.8915	18875.8444	18475.1232	18036.4718	17561.7401	17052.8634
6 δ	.0011	.0011	.0011	.0011	.0011	.0011	.0011	.0011	.0011	.0011	.0011	.0011	.0011
7 BO	18230.7945	18185.2662	18094.4443	17958.7944	17779.0049	17555.9765	17290.8076	16984.7781	16639.3314	16256.0546	15836.6588	15382.9599	14896.8586
7 AD	5.1086	5.1162	5.1312	5.1533	5.1824	5.2180	5.2596	5.3066	5.3584	5.4143	5.4734	5.5350	5.5982
7 NA	1.8823	1.8829	1.8842	1.8861	1.8887	1.8919	1.8957	1.9003	1.9055	1.9115	1.9181	1.9256	1.9338
7 α^2	-.5925	-.5932	-.5947	-.5969	-.5997	-.6032	-.6072	-.6118	-.6167	-.6221	-.6277	-.6335	-.6394
7 α^3	-.1052	-.1048	-.1041	-.1030	-.1015	-.0997	-.0975	-.0951	-.0923	-.0893	-.0860	-.0826	-.0789
7 α^4	-.0009	-.0009	-.0009	-.0009	-.0009	-.0009	-.0009	-.0008	-.0008	-.0008	-.0008	-.0007	-.0007
7 Total	18237.0868	18191.5664	18100.7600	17965.1331	17785.3739	17562.3826	17297.2573	16991.2773	16645.8854	16262.6681	15843.3359	15389.7037	14903.6716
7 δ	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012
8 BO	16043.0944	15999.5873	15912.8035	15783.1997	15611.4516	15398.4431	15145.2524	14853.1361	14523.5118	14157.9389	13758.0995	13325.7799	12862.8517
8 AD	5.1753	5.1814	5.1936	5.2116	5.2351	5.2637	5.2970	5.3343	5.3750	5.4184	5.4638	5.5103	5.5570
8 NA	2.0556	2.0560	2.0570	2.0584	2.0604	2.0628	2.0657	2.0692	2.0732	2.0778	2.0828	2.0885	2.0947
8 α^2	-.5894	-.5901	-.5914	-.5933	-.5958	-.5988	-.6023	-.6062	-.6105	-.6150	-.6197	-.6245	-.6294
8 α^3	-.0945	-.0941	-.0934	-.0923	-.0909	-.0892	-.0871	-.0848	-.0822	-.0793	-.0762	-.0729	-.0694
8 α^4	-.0008	-.0008	-.0008	-.0008	-.0008	-.0008	-.0008	-.0008	-.0007	-.0007	-.0007	-.0007	-.0006
8 Total	16049.6405	16006.1398	15919.3685	15789.7833	15618.0596	15405.0808	15151.9249	14859.8479	14530.2666	14164.7400	13764.9495	13332.6805	12869.8040
8 δ	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012
9 BO	13970.2613	13928.8029	13846.1125	13722.6391	13559.0470	13356.2055	13115.1752	12837.1928	12523.6538	12176.0945	11796.1738	11385.6547	10946.3872
9 AD	5.1694	5.1741	5.1833	5.1970	5.2147	5.2361	5.2607	5.2879	5.3172	5.3477	5.3788	5.4096	5.4394
9 NA	2.2094	2.2097	2.2103	2.2112	2.2124	2.2140	2.2158	2.2180	2.2205	2.2234	2.2265	2.2300	2.2339
9 α^2	-.5828	-.5834	-.5845	-.5861	-.5882	-.5907	-.5936	-.5969	-.6004	-.6040	-.6078	-.6116	-.6153
9 α^3	-.0842	-.0839	-.0832	-.0822	-.0809	-.0792	-.0773	-.0751	-.0726	-.0699	-.0669	-.0638	-.0605
9 α^4	-.0007	-.0007	-.0007	-.0007	-.0007	-.0007	-.0007	-.0007	-.0007	-.0006	-.0006	-.0006	-.0006
9 Total	13976.9722	13935.5186	13852.8377	13729.3782	13565.8044	13362.9849	13121.9801	12844.0261	12530.5179	12182.9911	11803.1038	11392.6185	10953.3842
9 δ	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013
10 BO	12014.0982	11974.7304	11896.2180	11779.0034	11623.7414	11431.2891	11202.6926	10939.1727	10642.1078	10313.0169	9953.5423	9565.4320	9150.5235
10 AD	5.0895	5.0927	5.0990	5.1082	5.1199	5.1339	5.1496	5.1664	5.1839	5.2011	5.2175	5.2323	5.2445
10 NA	2.3417	2.3418	2.3420	2.3422	2.3426	2.3431	2.3436	2.3442	2.3449	2.3456	2.3464	2.3472	2.3479
10 α^2	-.5723	-.5727	-.5736	-.5749	-.5766	-.5786	-.5809	-.5834	-.5861	-.5888	-.5915	-.5941	-.5966
10 α^3	-.0746	-.0742	-.0736	-.0726	-.0714	-.0698	-.0680	-.0659	-.0635	-.0610	-.0582	-.0552	-.0521
10 α^4	-.0007	-.0007	-.0007	-.0007	-.0006	-.0006	-.0006	-.0006	-.0006	-.0006	-.0005	-.0005	-.0005
10 Total	12020.8819	11981.5173	11903.0111	11785.8056	11630.5553	11438.1169	11209.5363	10946.0334	10648.9863	10319.9134	9960.4560	9572.3616	9157.4667
10 δ	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0013	.0013	.0013	.0013	.0013
11 BO	10177.2539	10140.0361	10065.8209	9955.0458	9808.3583	9626.6064	9410.8260	9162.2266	8882.1758	8572.1828	8233.8817	7869.0159	7479.4229
11 AD	4.9353	4.9370	4.9402	4.9448	4.9504	4.9568	4.9633	4.9696	4.9750	4.9787	4.9801	4.9785	4.9728
11 NA	2.4495	2.4494	2.4490	2.4492	2.4477	2.4467	2.4455	2.4441	2.4424	2.4404	2.4380	2.4352	2.4319
11 α^2	-.5574	-.5578	-.5584	-.5594	-.5606	-.5621	-.5637	-.5654	-.5672	-.5689	-.5705	-.5718	-.5729
11 α^3	-.0654	-.0651	-.0645	-.0636	-.0624	-.0609	-.0592	-.0572	-.0550	-.0526	-.0500	-.0472	-.0442
11 α^4	-.0006	-.0006	-.0006	-.0006	-.0006	-.0006	-.0005	-.0005	-.0005	-.0005	-.0005	-.0005	-.0004
11 Total	10184.0153	10146.7989	10072.5866	9961.8154	9815.1328	9633.3864	9417.6115	9169.0172	8888.9705	8578.9799	8240.6789	7875.8101	7486.2101
11 δ	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014

Components of dissociation energy of D₂, cont.

$v \setminus J$	0	1	2	3	4	5	6	7	8	9	10	11	12
12 BO	8463.3619	8428.3740	8358.6170	8254.5253	8116.7419	7946.1099	7743.6598	7510.5966	7248.2850	6958.2337	6642.0806	6301.5785	5938.5811
12 AD	4.7075	4.7076	4.7077	4.7076	4.7071	4.7058	4.7032	4.6988	4.6919	4.6820	4.6683	4.6500	4.6263
12 NA	2.5285	2.5279	2.5268	2.5252	2.5230	2.5202	2.5167	2.5125	2.5077	2.5019	2.4953	2.4876	2.4787
12 α^2	-5379	-5382	-5386	-5392	-5399	-5407	-5416	-5425	-5432	-5438	-5441	-5440	-5435
12 α^3	-0568	-0565	-0559	-0550	-0539	-0525	-0509	-0490	-0469	-0447	-0422	-0396	-0369
12 α^4	-0005	-0005	-0005	-0005	-0005	-0005	-0005	-0005	-0004	-0004	-0004	-0004	-0004
12 Total	8470.0026	8435.0143	8365.2566	8261.1634	8123.3777	7952.7421	7750.2867	7517.2160	7254.8939	6964.8287	6648.6575	6308.1321	5945.1055
12 δ	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014
13 BO	6877.2164	6844.5640	6779.4775	6682.3905	6553.9458	6394.9864	6206.5446	5989.8295	5746.2134	5477.2179	5184.5006	4869.8427	4535.1382
13 AD	4.4085	4.4070	4.4041	4.3994	4.3928	4.3837	4.3720	4.3569	4.3379	4.3144	4.2856	4.2508	4.2091
13 NA	2.5721	2.5711	2.5691	2.5660	2.5618	2.5564	2.5499	2.5421	2.5328	2.5221	2.5096	2.4952	2.4786
13 α^2	-5132	-5133	-5135	-5136	-5138	-5140	-5140	-5139	-5135	-5127	-5116	-5099	-5075
13 α^3	-0486	-0483	-0478	-0470	-0459	-0446	-0431	-0413	-0394	-0373	-0350	-0326	-0300
13 α^4	-0005	-0005	-0004	-0004	-0004	-0004	-0004	-0004	-0004	-0004	-0003	-0003	-0003
13 Total	6883.6348	6850.9801	6785.8889	6688.7948	6560.3401	6401.3676	6212.9089	5996.1728	5752.5308	5483.5039	5190.7489	4876.0459	4541.2881
13 δ	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014
14 BO	5424.9997	5394.8201	5334.6803	5245.0169	5126.4772	4979.9117	4806.3636	4607.0583	4383.3908	4136.9141	3869.3284	3582.4722	3278.3150
14 AD	4.0428	4.0398	4.0338	4.0246	4.0119	3.9954	3.9746	3.9490	3.9181	3.8811	3.8375	3.7863	3.7267
14 NA	2.5720	2.5703	2.5670	2.5619	2.5550	2.5463	2.5355	2.5227	2.5075	2.4898	2.4692	2.4456	2.4183
14 α^2	-4827	-4826	-4824	-4821	-4817	-4810	-4801	-4789	-4772	-4750	-4721	-4685	-4640
14 α^3	-0409	-0407	-0401	-0394	-0384	-0372	-0358	-0342	-0323	-0304	-0283	-0260	-0237
14 α^4	-0004	-0004	-0004	-0004	-0004	-0004	-0004	-0003	-0003	-0003	-0003	-0003	-0003
14 Total	5431.0906	5400.9066	5340.7582	5251.0815	5132.5237	4985.9347	4812.3575	4613.0166	4389.3065	4142.7793	3875.1345	3588.2093	3283.9721
14 δ	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0013
15 BO	4114.5748	4087.0460	4032.2117	3950.5157	3842.6174	3709.3849	3551.8863	3371.3804	3169.3070	2947.2788	2707.0741	2450.6331	2180.0588
15 AD	3.6167	3.6123	3.6033	3.5897	3.5711	3.5472	3.5176	3.4818	3.4391	3.3889	3.3305	3.2629	3.1850
15 NA	2.5168	2.5143	2.5092	2.5014	2.4909	2.4775	2.4610	2.4412	2.4179	2.3907	2.3590	2.3224	2.2802
15 α^2	-4454	-4452	-4447	-4438	-4427	-4411	-4391	-4365	-4333	-4293	-4244	-4185	-4114
15 α^3	-0337	-0334	-0330	-0323	-0314	-0303	-0289	-0274	-0258	-0240	-0220	-0200	-0179
15 α^4	-0003	-0003	-0003	-0003	-0003	-0003	-0003	-0003	-0003	-0003	-0002	-0002	-0002
15 Total	4120.2289	4092.6936	4037.8462	3956.1303	3848.2050	3714.9379	3557.3966	3376.8392	3174.7048	2952.6049	2712.3169	2455.7797	2185.0946
15 δ	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0014	.0013	.0013	.0013	.0013	.0013
16 BO	2955.8684	2931.2223	2882.1605	2809.1411	2712.8468	2594.1785	2454.2498	2294.3801	2116.0893	1921.0948	1711.3124	1488.8629	1256.0886
16 AD	3.1380	3.1321	3.1203	3.1023	3.0780	3.0469	3.0085	2.9625	2.9079	2.8442	2.7703	2.6851	2.5871
16 NA	2.3923	2.3886	2.3810	2.3696	2.3541	2.3343	2.3100	2.2808	2.2462	2.2056	2.1584	2.1036	2.0401
16 α^2	-4005	-4001	-3991	-3977	-3956	-3930	-3896	-3855	-3804	-3742	-3668	-3580	-3475
16 α^3	-0269	-0267	-0263	-0256	-0248	-0238	-0226	-0212	-0197	-0180	-0163	-0145	-0125
16 α^4	-0003	-0003	-0003	-0003	-0003	-0002	-0002	-0002	-0002	-0002	-0002	-0002	-0001
16 Total	2960.9710	2936.3160	2887.2361	2814.1894	2717.8581	2599.1427	2459.1559	2299.2164	2120.8431	1925.7521	1715.8578	1493.2790	1260.3555
16 δ	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0012	.0012	.0012	.0012	.0011
17 BO	1961.3767	1939.9186	1897.2449	1833.8397	1750.4259	1647.9625	1527.6426	1390.8921	1239.3726	1074.9901	899.9132	716.6069	527.8945
17 AD	2.6140	2.6067	2.5920	2.5696	2.5393	2.5006	2.4531	2.3960	2.3284	2.2492	2.1570	2.0499	1.9251
17 NA	2.1802	2.1748	2.1640	2.1476	2.1253	2.0969	2.0618	2.0195	1.9693	1.9101	1.8407	1.7593	1.6634
17 α^2	-3467	-3460	-3445	-3423	-3392	-3352	-3302	-3241	-3166	-3077	-2970	-2842	-2689
17 α^3	-0206	-0204	-0200	-0194	-0187	-0177	-0167	-0154	-0141	-0126	-0110	-0094	-0077
17 α^4	-0002	-0002	-0002	-0002	-0002	-0002	-0002	-0002	-0002	-0001	-0001	-0001	-0001
17 Total	1965.8035	1944.3335	1901.6362	1838.1950	1754.7324	1652.2069	1531.8105	1394.9679	1243.3394	1078.8290	903.6027	720.1223	531.2062
17 δ	.0012	.0012	.0012	.0012	.0012	.0012	.0011	.0011	.0011	.0011	.0010	.0010	.0009

Components of dissociation energy of D₂, cont.

v\J	0	1	2	3	4	5	6	7	8	9	10	11	12
18 BO	1146.8377	1128.9787	1093.5250	1041.0054	972.2143	888.2144	790.3427	680.2231	559.7902	431.3322	297.5703	161.8141	28.3073
18 AD	2.0490	2.0400	2.0220	1.9945	1.9573	1.9097	1.8509	1.7798	1.6948	1.5939	1.4738	1.3294	1.1507
18 NA	1.8591	1.8515	1.8362	1.8130	1.7814	1.7409	1.6907	1.6299	1.5569	1.4698	1.3655	1.2394	1.0822
18 α^2	-.2822	-.2812	-.2791	-.2759	-.2715	-.2658	-.2587	-.2499	-.2392	-.2263	-.2106	-.1914	-.1672
18 α^3	-.0147	-.0146	-.0142	-.0137	-.0130	-.0122	-.0112	-.0101	-.0089	-.0076	-.0063	-.0049	-.0034
18 α^4	-.0002	-.0002	-.0002	-.0001	-.0001	-.0001	-.0001	-.0001	-.0001	-.0001	-.0001	-.0001	-.0001
18 Total	1150.4486	1132.5743	1097.0897	1044.5231	975.6684	891.5869	793.6143	683.3726	562.7936	434.1618	300.1927	164.1866	30.3695
18 δ	.0010	.0010	.0010	.0010	.0010	.0010	.0009	.0009	.0009	.0008	.0008	.0007	.0006
19 BO	532.1061	518.4250	491.3676	451.5479	399.8985	337.6900	266.5650	188.6031	106.4515	23.6339			
19 AD	1.4389	1.4277	1.4051	1.3706	1.3235	1.2626	1.1862	1.0915	.9741	.8243			
19 NA	1.4039	1.3932	1.3716	1.3388	1.2938	1.2356	1.1626	1.0721	.9597	.8164			
19 α^2	-.2051	-.2037	-.2007	-.1962	-.1899	-.1818	-.1715	-.1586	-.1423	-.1214			
19 α^3	-.0093	-.0092	-.0089	-.0084	-.0078	-.0071	-.0062	-.0053	-.0042	-.0031			
19 α^4	-.0001	-.0001	-.0001	-.0001	-.0001	-.0001	-.0001	-.0001	-.0001	.0000			
19 Total	534.7343	521.0330	493.9347	454.0526	402.3180	339.9993	268.7360	190.6028	108.2387	25.1501			
19 δ	.0008	.0008	.0008	.0007	.0007	.0007	.0006	.0006	.0005	.0005			
20 BO	142.0246	133.4163	116.6008	92.4041	62.1411	27.7727							
20 AD	.7671	.7521	.7215	.6741	.6071	.5151							
20 NA	.7896	.7742	.7428	.6941	.6255	.5316							
20 α^2	-.1127	-.1106	-.1063	-.0995	-.0900	-.0769							
20 α^3	-.0043	-.0042	-.0039	-.0035	-.0030	-.0023							
20 α^4	-.0001	.0000	.0000	.0000	.0000	.0000							
20 Total	143.4642	134.8278	117.9548	93.6691	63.2806	28.7401							
20 δ	.0004	.0004	.0004	.0004	.0003	.0003							
21 BO	1.5444	— ^a											
21 AD	.0633	.0232											
21 NA	.0670	.0310											
21 α^2	-.0102	-.0049											
21 α^3	-.0002	-.0001											
21 α^4	.0000	.0000											
21 Total	1.6642	.0491											
21 δ	.0000	.0000											

^a Does not exist in this approximation.

Components of dissociation energy of D₂, cont.

v\J	13	14	15	16	17	18	19	20	21	22	23	24	25
0 BO	31648.1896	30919.6643	30153.8636	29353.2065	28520.1105	27656.9774	26766.1800	25850.0516	24910.8764	23950.8826	22972.2363	21977.0372	20967.3160
0 AD	4.0931	4.2720	4.4573	4.6482	4.8435	5.0424	5.2439	5.4470	5.6509	5.8547	6.0576	6.2586	6.4569
0 NA	.3106	.3341	.3593	.3861	.4145	.4446	.4763	.5096	.5446	.5811	.6193	.6591	.7006
0 α^2	-.6296	-.6426	-.6559	-.6693	-.6828	-.6964	-.7098	-.7231	-.7362	-.7490	-.7614	-.7734	-.7850
0 α^3	-.1563	-.1507	-.1448	-.1388	-.1326	-.1263	-.1199	-.1134	-.1069	-.1004	-.0938	-.0872	-.0807
0 α^4	-.0013	-.0013	-.0012	-.0012	-.0011	-.0011	-.0010	-.0010	-.0009	-.0009	-.0008	-.0008	-.0007
0 Total	31651.8060	30923.4758	30157.8783	29357.4315	28524.5520	27661.6407	26771.0694	25855.1707	24916.2279	23956.4683	22978.0571	21983.0935	20973.6071
0 δ	.0008	.0007	.0007	.0007	.0007	.0007	.0007	.0007	.0007	.0007	.0007	.0007	.0007
1 BO	28837.4015	28135.7076	27398.2684	26627.4371	25825.5661	24994.9924	24138.0260	23256.9391	22353.9576	21431.2546	20490.9446	19535.0802	18565.6491
1 AD	4.5136	4.6776	4.8471	5.0212	5.1989	5.3793	5.5615	5.7444	5.9273	6.1092	6.2893	6.4667	6.6406
1 NA	.5993	.6206	.6434	.6678	.6937	.7211	.7501	.7806	.8127	.8464	.8816	.9185	.9569
1 α^2	-.6407	-.6527	-.6650	-.6774	-.6898	-.7022	-.7145	-.7266	-.7385	-.7500	-.7611	-.7718	-.7820
1 α^3	-.1428	-.1374	-.1318	-.1260	-.1201	-.1141	-.1080	-.1019	-.0956	-.0894	-.0832	-.0769	-.0707
1 α^4	-.0012	-.0012	-.0011	-.0011	-.0010	-.0010	-.0009	-.0009	-.0008	-.0008	-.0007	-.0007	-.0007
1 Total	28841.7298	28140.2145	27402.9610	26632.3216	25830.6476	25000.2755	24143.5140	23262.6347	22359.8626	21437.3699	20497.2704	19541.6158	18572.3932
1 δ	.0008	.0008	.0008	.0008	.0007	.0007	.0007	.0007	.0007	.0007	.0007	.0007	.0007
2 BO	26142.8111	25467.5564	24758.0795	24016.6728	23245.6284	22447.2242	21623.7128	20777.3113	19910.1940	19024.4855	18122.2562	17205.5192	16276.2279
2 AD	4.8722	5.0208	5.1741	5.3310	5.4906	5.6520	5.8142	5.9762	6.1373	6.2964	6.4528	6.6055	6.7537
2 NA	.8685	.8878	.9084	.9305	.9540	.9790	1.0055	1.0335	1.0629	1.0939	1.1264	1.1605	1.1961
2 α^2	-.6490	-.6601	-.6714	-.6827	-.6941	-.7053	-.7164	-.7273	-.7379	-.7481	-.7579	-.7672	-.7759
2 α^3	-.1299	-.1248	-.1194	-.1139	-.1083	-.1026	-.0968	-.0909	-.0850	-.0791	-.0731	-.0672	-.0613
2 α^4	-.0011	-.0011	-.0010	-.0010	-.0009	-.0009	-.0009	-.0008	-.0008	-.0007	-.0007	-.0006	-.0006
2 Total	26147.7718	25472.6790	24763.3702	24022.1367	23251.2697	22453.0464	21629.7184	20783.5020	19916.5706	19031.0480	18129.0038	17212.4502	16283.3399
2 δ	.0008	.0008	.0008	.0008	.0008	.0008	.0008	.0008	.0008	.0008	.0008	.0008	.0008
3 BO	23562.9810	22913.8768	22232.0753	21519.8130	20779.3270	20012.8419	19222.5590	18410.6475	17579.2366	16730.4106	15866.2041	14988.5997	14099.5266
3 AD	5.1668	5.2998	5.4363	5.5755	5.7163	5.8580	5.9994	6.1397	6.2781	6.4134	6.5450	6.6718	6.7930
3 NA	1.1192	1.1365	1.1551	1.1751	1.1964	1.2191	1.2432	1.2687	1.2957	1.3241	1.3539	1.3852	1.4181
3 α^2	-.6545	-.6646	-.6749	-.6851	-.6954	-.7055	-.7153	-.7250	-.7342	-.7430	-.7514	-.7592	-.7665
3 α^3	-.1177	-.1128	-.1077	-.1025	-.0971	-.0917	-.0861	-.0806	-.0749	-.0693	-.0637	-.0581	-.0525
3 α^4	-.0010	-.0010	-.0009	-.0009	-.0009	-.0008	-.0008	-.0008	-.0007	-.0006	-.0006	-.0006	-.0005
3 Total	23568.4939	22919.5347	22237.8832	21525.7751	20785.4464	20019.1210	19228.9994	18417.2497	17586.0005	16737.3351	15873.2873	14995.8389	14106.9182
3 δ	.0009	.0009	.0009	.0009	.0009	.0009	.0009	.0009	.0009	.0009	.0009	.0009	.0009
4 BO	21096.9577	20473.8275	19819.5362	19136.2699	18426.2165	17691.5537	16934.4388	16157.0001	15361.3308	14549.4833	13723.4666	12885.2440	12036.7325
4 AD	5.3956	5.5123	5.6314	5.7522	5.8736	5.9947	6.1145	6.2321	6.3466	6.4570	6.5624	6.6619	6.7545
4 NA	1.3519	1.3674	1.3841	1.4020	1.4212	1.4417	1.4634	1.4865	1.5110	1.5367	1.5639	1.5924	1.6223
4 α^2	-.6570	-.6662	-.6754	-.6845	-.6936	-.7025	-.7111	-.7194	-.7272	-.7346	-.7415	-.7477	-.7533
4 α^3	-.1061	-.1015	-.0966	-.0916	-.0865	-.0814	-.0761	-.0708	-.0655	-.0601	-.0548	-.0495	-.0442
4 α^4	-.0009	-.0009	-.0008	-.0008	-.0008	-.0007	-.0007	-.0006	-.0006	-.0006	-.0005	-.0005	-.0004
4 Total	21102.9411	20479.9386	19825.7789	19142.6472	18432.7304	17698.2055	16941.2288	16163.9280	15368.3950	14556.6818	13730.7962	12892.7006	12044.3114
4 δ	.0009	.0009	.0009	.0009	.0009	.0009	.0010	.0010	.0010	.0010	.0010	.0010	.0010
5 BO	18744.3086	18147.0996	17520.2880	16866.0157	16186.4281	15483.6635	14759.8430	14017.0634	13257.3912	12482.8585	11695.4608	10897.1558	10089.8643
5 AD	5.5561	5.6558	5.7570	5.8585	5.9596	6.0592	6.1564	6.2502	6.3397	6.4238	6.5017	6.5723	6.6346
5 NA	1.5669	1.5806	1.5953	1.6112	1.6283	1.6465	1.6659	1.6865	1.7083	1.7313	1.7554	1.7808	1.8074
5 α^2	-.6565	-.6646	-.6727	-.6807	-.6885	-.6961	-.7034	-.7103	-.7166	-.7225	-.7277	-.7322	-.7360
5 α^3	-.0952	-.0907	-.0861	-.0814	-.0765	-.0716	-.0666	-.0616	-.0566	-.0515	-.0465	-.0415	-.0365
5 α^4	-.0008	-.0008	-.0008	-.0007	-.0007	-.0007	-.0006	-.0006	-.0005	-.0005	-.0005	-.0004	-.0004
5 Total	18750.6790	18153.5798	17526.8807	16872.7226	16193.2502	15490.6008	14766.8946	14024.2276	13264.6654	12490.2391	11702.9433	10904.7348	10097.5334
5 δ	.0010	.0010	.0010	.0010	.0010	.0010	.0010	.0010	.0010	.0011	.0011	.0011	.0011

Components of dissociation energy of D₂, cont.

$v \setminus J$	13	14	15	16	17	18	19	20	21	22	23	24	25
6 BO	16505.1722	15933.9699	15334.7593	14709.6448	14060.7374	13390.1444	12699.9608	11992.2624	11269.1016	10532.5040	9784.4671	9026.9614	8261.9325
6 AD	5.6457	5.7279	5.8102	5.8918	5.9716	6.0487	6.1222	6.1910	6.2542	6.3107	6.3595	6.3995	6.4297
6 NA	1.7641	1.7759	1.7886	1.8024	1.8172	1.8329	1.8497	1.8675	1.8864	1.9062	1.9269	1.9486	1.9711
6 α^2	-6527	-6597	-6666	-6734	-6800	-6862	-6920	-6974	-7021	-7063	-7097	-7123	-7141
6 α^3	-0848	-0806	-0762	-0717	-0671	-0624	-0577	-0530	-0482	-0434	-0387	-0340	-0293
6 α^4	-0008	-0007	-0007	-0007	-0006	-0006	-0005	-0005	-0005	-0004	-0004	-0004	-0003
6 Total	16511.8438	15940.7327	15341.6146	14716.5931	14067.7784	13397.2768	12707.1824	11999.5701	11276.4914	10539.9706	9792.0047	9034.5628	8269.5895
6 δ	.0011	.0011	.0011	.0011	.0011	.0011	.0011	.0011	.0011	.0011	.0011	.0011	.0012
7 BO	14380.3229	13835.3711	13264.0568	12668.4554	12050.6530	11412.7365	10756.7866	10084.8724	9399.0474	8701.3489	7993.7978	7278.4022	6557.1621
7 AD	5.6622	5.7260	5.7887	5.8494	5.9070	5.9606	6.0092	6.0518	6.0873	6.1147	6.1329	6.1406	6.1368
7 NA	1.9427	1.9525	1.9631	1.9744	1.9866	1.9996	2.0133	2.0278	2.0430	2.0588	2.0752	2.0921	2.1093
7 α^2	-6453	-6512	-6569	-6624	-6675	-6723	-6765	-6802	-6832	-6855	-6869	-6874	-6869
7 α^3	-0750	-0710	-0668	-0626	-0582	-0538	-0494	-0449	-0404	-0359	-0315	-0271	-0227
7 α^4	-0007	-0006	-0006	-0006	-0005	-0005	-0005	-0004	-0004	-0004	-0003	-0003	-0003
7 Total	14387.2068	13842.3268	13271.0842	12675.5536	12057.8202	11419.9700	10764.0828	10092.2265	9406.4537	8708.8007	8001.2872	7285.9202	6564.6983
7 δ	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012
8 BO	12371.2548	11852.9818	11310.0633	10744.5563	10158.5333	9554.0748	8933.2629	8298.1773	7650.8943	6993.4875	6328.0310	5656.6060	4981.3107
8 AD	5.6032	5.6479	5.6902	5.7291	5.7635	5.7926	5.8153	5.8305	5.8371	5.8340	5.8199	5.7937	5.7539
8 NA	2.1015	2.1089	2.1168	2.1253	2.1343	2.1438	2.1537	2.1640	2.1745	2.1852	2.1958	2.2061	2.2158
8 α^2	-6342	-6388	-6432	-6472	-6509	-6540	-6565	-6583	-6593	-6595	-6587	-6574	-6535
8 α^3	-0657	-0619	-0580	-0540	-0499	-0457	-0416	-0373	-0331	-0289	-0248	-0207	-0167
8 α^4	-0006	-0006	-0005	-0005	-0005	-0004	-0004	-0004	-0003	-0003	-0003	-0002	-0002
8 Total	12378.2591	11860.0373	11317.1686	10751.7089	10165.7299	9561.3111	8940.5334	8305.4757	7658.2131	7000.8179	6335.3630	5663.9281	4988.6099
8 δ	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0013	.0013	.0013	.0013	.0013
9 BO	10480.2917	9989.3448	9475.5658	8941.0062	8387.7407	7817.8607	7233.4706	6636.6860	6029.6353	5414.4640	4793.3433	4168.4822	3542.1463
9 AD	5.4672	5.4921	5.5132	5.5294	5.5399	5.5435	5.5392	5.5259	5.5024	5.4674	5.4196	5.3576	5.2797
9 NA	2.2380	2.2424	2.2471	2.2519	2.2568	2.2618	2.2667	2.2713	2.2755	2.2790	2.2814	2.2824	2.2815
9 α^2	-6188	-6221	-6250	-6275	-6294	-6307	-6313	-6311	-6298	-6276	-6241	-6192	-6129
9 α^3	-0570	-0534	-0497	-0460	-0421	-0382	-0343	-0303	-0264	-0225	-0187	-0149	-0112
9 α^4	-0005	-0005	-0005	-0004	-0004	-0004	-0004	-0003	-0003	-0003	-0002	-0002	-0002
9 Total	10487.3206	9996.4033	9482.6509	8948.1137	8394.8655	7824.9967	7240.6106	6643.8215	6036.7566	5421.5600	4800.4013	4175.4879	3549.0832
9 δ	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013
10 BO	8710.7289	8248.0215	7764.4241	7262.0005	6742.8480	6209.0938	5662.8939	5106.4354	4541.9427	3971.6882	3398.0093	2823.3333	2250.2131
10 AD	5.2533	5.2579	5.2571	5.2501	5.2359	5.2132	5.1811	5.1383	5.0835	5.0153	4.9322	4.8323	4.7134
10 NA	2.3486	2.3491	2.3495	2.3494	2.3489	2.3478	2.3457	2.3425	2.3377	2.3310	2.3217	2.3090	2.2921
10 α^2	-5987	-6005	-6018	-6026	-6026	-6019	-6003	-5976	-5938	-5886	-5820	-5736	-5632
10 α^3	-0489	-0455	-0420	-0384	-0348	-0312	-0275	-0239	-0203	-0167	-0132	-0097	-0064
10 α^4	-0005	-0004	-0004	-0004	-0004	-0003	-0003	-0003	-0002	-0002	-0002	-0002	-0001
10 Total	8717.6828	8254.9820	7771.3865	7268.9587	6749.7950	6216.0214	5669.7926	5113.2944	4548.7496	3978.4290	3404.6679	2829.8912	2256.6490
10 δ	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0013
11 BO	7067.0208	6633.7968	6181.7978	5713.1238	5229.9242	4734.3976	4228.7956	3715.4317	3196.6961	2675.0790	2153.2050	1633.8858	1120.1985
11 AD	4.9623	4.9461	4.9231	4.8924	4.8528	4.8033	4.7426	4.6693	4.5821	4.4790	4.3581	4.2168	4.0519
11 NA	2.4280	2.4233	2.4177	2.4110	2.4030	2.3932	2.3814	2.3670	2.3493	2.3276	2.3009	2.2676	2.2260
11 α^2	-5735	-5736	-5731	-5719	-5698	-5667	-5625	-5569	-5499	-5412	-5306	-5176	-5019
11 α^3	-0412	-0380	-0348	-0315	-0281	-0247	-0214	-0180	-0147	-0114	-0082	-0052	-0022
11 α^4	-0004	-0004	-0004	-0003	-0003	-0003	-0002	-0002	-0002	-0002	-0001	-0001	-0001
11 Total	7073.7960	6640.5542	6188.5304	5719.8236	5236.5818	4741.0024	4235.3355	3721.8929	3203.0627	2681.3328	2159.3251	1639.8473	1125.9722
11 δ	.0014	.0014	.0014	.0014	.0014	.0013	.0013	.0013	.0013	.0013	.0013	.0013	.0012

Components of dissociation energy of D₂, cont.

$v \setminus J$	13	14	15	16	17	18	19	20	21	22	23	24	25
12 BO	5555.0327	5152.9583	4734.4578	4301.7034	3856.9403	3402.4939	2940.7819	2474.3359	2005.8348	1538.1552	1074.4514	618.2827	173.8344
12 AD	4.5964	4.5591	4.5137	4.4589	4.3937	4.3167	4.2265	4.1215	3.9997	3.8586	3.6950	3.5045	3.2800
12 NA	2.4685	2.4566	2.4429	2.4268	2.4081	2.3862	2.3603	2.3297	2.2930	2.2490	2.1954	2.1293	2.0463
12 α^2	-5424	-5406	-5380	-5344	-5298	-5238	-5165	-5074	-4964	-4832	-4671	-4477	-4240
12 α^3	-0340	-0311	-0281	-0250	-0219	-0188	-0157	-0127	-0097	-0068	-0040	-0013	.0012
12 α^4	-0003	-0003	-0003	-0003	-0002	-0002	-0002	-0002	-0001	-0001	-0001	-0001	.0000
12 Total	5561.5208	5159.4021	4740.8480	4308.0294	3863.1902	3408.6539	2946.8363	2480.2668	2011.6212	1543.7728	1079.8706	623.4674	178.7378
12 δ	.0014	.0014	.0014	.0014	.0013	.0013	.0013	.0013	.0013	.0013	.0012	.0012	.0011
13 BO	4182.3861	3813.6849	3431.2318	3037.3266	2634.3822	2224.9439	1811.7210	1397.6371	985.9096	580.1812	184.7502		
13 AD	4.1597	4.1014	4.0333	3.9541	3.8625	3.7569	3.6354	3.4956	3.3343	3.1470	2.9267		
13 NA	2.4596	2.4377	2.4124	2.3832	2.3493	2.3098	2.2634	2.2085	2.1428	2.0632	1.9647		
13 α^2	-5044	-5004	-4953	-4889	-4812	-4718	-4604	-4468	-4304	-4106	-3864		
13 α^3	-0274	-0247	-0219	-0191	-0163	-0135	-0107	-0080	-0053	-0028	-0004		
13 α^4	-0003	-0003	-0002	-0002	-0002	-0002	-0001	-0001	-0001	-0001	-0001		
13 Total	4188.4733	3819.6987	3437.1601	3043.1557	2640.0964	2230.5252	1817.1485	1402.8863	990.9510	584.9779	189.2547		
13 δ	.0014	.0014	.0013	.0013	.0013	.0013	.0013	.0012	.0012	.0011	.0011		
14 BO	2958.9550	2626.6208	2283.6796	1932.6552	1576.2577	1217.4335	859.4468	506.0169	161.5669				
14 AD	3.6577	3.5781	3.4868	3.3821	3.2622	3.1248	2.9666	2.7830	2.5667				
14 NA	2.3870	2.3510	2.3095	2.2613	2.2053	2.1395	2.0617	1.9683	1.8538				
14 α^2	-4585	-4517	-4436	-4337	-4220	-4080	-3912	-3709	-3461				
14 α^3	-0213	-0188	-0163	-0137	-0112	-0087	-0063	-0039	-0017				
14 α^4	-0002	-0002	-0002	-0002	-0001	-0001	-0001	-0001	-0001				
14 Total	2964.5198	2632.0792	2289.0159	1937.8510	1581.2918	1222.2810	864.0776	510.3933	165.6395				
14 δ	.0013	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0010				
15 BO	1897.6238	1605.7866	1307.2212	1004.8675	702.0170	402.4614	110.7743						
15 AD	3.0957	2.9936	2.8767	2.7426	2.5882	2.4085	2.1954						
15 NA	2.2315	2.1752	2.1099	2.0336	1.9437	1.8362	1.7047						
15 α^2	-4028	-3926	-3805	-3661	-3488	-3280	-3024						
15 α^3	-0157	-0134	-0112	-0089	-0067	-0046	-0025						
15 α^4	-0002	-0002	-0001	-0001	-0001	-0001	-0001						
15 Total	1902.5323	1610.5491	1311.8159	1009.2686	706.1933	406.3736	114.3694						
15 δ	.0012	.0012	.0012	.0011	.0011	.0010	.0009						
16 BO	1015.5833	770.2463	523.3762	278.8421	41.4288								
16 AD	2.4745	2.3448	2.1945	2.0181	1.8061								
16 NA	1.9661	1.8798	1.7778	1.6558	1.5055								
16 α^2	-3350	-3201	-3023	-2808	-2541								
16 α^3	-0106	-0086	-0067	-0047	-0028								
16 α^4	-0001	-0001	-0001	-0001	-0001								
16 Total	1019.6781	774.1419	527.0394	282.2304	44.4834								
16 δ	.0011	.0010	.0010	.0009	.0008								
17 BO	337.0689	148.1118											
17 AD	1.7786	1.6038											
17 NA	1.5496	1.4119											
17 α^2	-2505	-2280											
17 α^3	-0060	-0043											
17 α^4	-0001	-0001											
17 Total	340.1405	150.8950											
17 δ	.0009	.0008											

Components of dissociation energy of D₂, cont.

$v \setminus J$	26	27	28	29	30	31	32	33	34	35	36	37	38
0 BO	19945.0321	18912.0732	17870.2556	16821.3251	15766.9589	14708.7679	13648.2998	12587.0428	11526.4295	10467.8416	9412.6148	8362.0445	7317.3923
0 AD	6.6519	7.8427	7.0285	7.2086	7.3822	7.5487	7.7072	7.8571	7.9975	8.1276	8.2466	8.3537	8.4478
0 NA	.7437	.7884	.8348	.8829	.9326	.9841	1.0372	1.0921	1.1487	1.2070	1.2672	1.3291	1.3928
0 α^2	-.7960	-.8066	-.8165	-.8258	-.8344	-.8423	-.8495	-.8559	-.8615	-.8663	-.8702	-.8731	-.8752
0 α^3	-.0742	-.0678	-.0614	-.0551	-.0489	-.0428	-.0369	-.0310	-.0252	-.0196	-.0141	-.0088	-.0036
0 α^4	-.0007	-.0006	-.0006	-.0005	-.0005	-.0004	-.0004	-.0004	-.0003	-.0003	-.0002	-.0002	-.0001
0 Total	19951.5567	18918.8293	17877.2405	16828.5352	15774.3900	14716.4151	13656.1575	12595.1047	11534.6886	10476.2901	9421.2440	8370.8452	7326.3541
0 δ	.0007	.0007	.0007	.0007	.0007	.0007	.0008	.0008	.0008	.0008	.0008	.0009	.0009
1 BO	17584.5734	16593.7086	15594.8446	14589.7075	13579.9615	12567.2120	11553.0096	10538.8541	9526.1997	8516.4610	7511.0194	6511.2308	5518.4347
1 AD	6.8101	6.9745	7.1330	7.2847	7.4289	7.5647	7.6913	7.8079	7.9135	8.0072	8.0880	8.1548	8.2064
1 NA	.9970	1.0387	1.0820	1.1270	1.1737	1.2221	1.2721	1.3239	1.3773	1.4325	1.4894	1.5480	1.6083
1 α^2	-.7917	-.8007	-.8092	-.8169	-.8240	-.8302	-.8357	-.8403	-.8441	-.8469	-.8488	-.8496	-.8493
1 α^3	-.0646	-.0585	-.0524	-.0465	-.0406	-.0348	-.0292	-.0236	-.0182	-.0129	-.0078	-.0028	.0021
1 α^4	-.0006	-.0006	-.0005	-.0005	-.0004	-.0004	-.0003	-.0003	-.0002	-.0002	-.0002	-.0001	-.0001
1 Total	17591.5237	16600.8620	15602.1976	14597.2554	13587.6991	12575.1334	11561.1078	10547.1215	9534.6279	8525.0406	7519.7400	6520.0811	5527.4021
1 δ	.0008	.0008	.0008	.0008	.0008	.0008	.0008	.0009	.0009	.0009	.0009	.0010	.0010
2 BO	15336.2758	14387.4959	13431.6630	12470.4951	11505.6568	10538.7632	9571.3848	8605.0531	7641.2673	6681.5026	5727.2195	4779.8752	3840.9379
2 AD	6.8966	7.0332	7.1628	7.2844	7.3971	7.5001	7.5924	7.6731	7.7409	7.7948	7.8336	7.8558	7.8600
2 NA	1.2333	1.2721	1.3124	1.3544	1.3980	1.4432	1.4900	1.5385	1.5885	1.6401	1.6933	1.7479	1.8040
2 α^2	-.7841	-.7916	-.7985	-.8046	-.8099	-.8144	-.8180	-.8206	-.8223	-.8229	-.8224	-.8207	-.8177
2 α^3	-.0555	-.0497	-.0440	-.0383	-.0328	-.0274	-.0220	-.0168	-.0118	-.0068	-.0020	.0026	.0071
2 α^4	-.0005	-.0005	-.0004	-.0004	-.0004	-.0003	-.0003	-.0002	-.0002	-.0002	-.0001	-.0001	.0000
2 Total	15343.5655	14394.9594	13439.2953	12478.2906	11513.6089	10546.8645	9579.6270	8613.4269	7649.7624	6690.1076	5735.9218	4788.6608	3849.7912
2 δ	.0008	.0009	.0009	.0009	.0009	.0009	.0009	.0010	.0010	.0010	.0010	.0011	.0011
3 BO	13200.8605	12294.4244	11381.9911	10465.2862	9545.9923	8625.7547	7706.1877	6788.8826	5875.4182	4967.3721	4066.3359	3173.9343	2291.8498
3 AD	6.9076	7.0148	7.1136	7.2030	7.2821	7.3498	7.4051	7.4466	7.4732	7.4834	7.4756	7.4480	7.3983
3 NA	1.4524	1.4882	1.5255	1.5643	1.6046	1.6464	1.6896	1.7342	1.7801	1.8273	1.8756	1.9248	1.9748
3 α^2	-.7730	-.7789	-.7840	-.7883	-.7918	-.7942	-.7957	-.7962	-.7954	-.7935	-.7902	-.7854	-.7790
3 α^3	-.0470	-.0415	-.0361	-.0308	-.0256	-.0205	-.0155	-.0106	-.0059	-.0013	.0032	.0074	.0115
3 α^4	-.0005	-.0004	-.0004	-.0003	-.0003	-.0003	-.0002	-.0002	-.0001	-.0001	-.0001	.0000	.0000
3 Total	13208.4000	12302.1065	11389.8096	10473.2340	9554.0614	8633.9359	7714.4708	6797.2565	5883.8701	4975.8879	4074.9000	3182.5291	2300.4554
3 δ	.0009	.0009	.0010	.0010	.0010	.0010	.0010	.0010	.0011	.0011	.0011	.0011	.0012
4 BO	11179.8040	10316.2870	9447.9704	8576.6080	7703.9249	6831.6252	5961.4019	5094.9491	4233.9777	3380.2353	2535.5322	1701.7765	881.0243
4 AD	6.8392	6.9151	6.9810	7.0360	7.0789	7.1085	7.1235	7.1224	7.1036	7.0653	7.0052	6.9208	6.8088
4 NA	1.6535	1.6861	1.7200	1.7552	1.7917	1.8293	1.8680	1.9076	1.9481	1.9891	2.0303	2.0715	2.1119
4 α^2	-.7582	-.7622	-.7654	-.7677	-.7690	-.7693	-.7683	-.7661	-.7626	-.7575	-.7507	-.7421	-.7313
4 α^3	-.0390	-.0338	-.0288	-.0238	-.0189	-.0141	-.0095	-.0049	-.0006	.0036	.0077	.0116	.0152
4 α^4	-.0004	-.0004	-.0003	-.0003	-.0002	-.0002	-.0002	-.0001	-.0001	-.0001	.0000	.0000	.0000
4 Total	11187.4992	10324.0917	9455.8769	8584.6075	7712.0074	6839.7795	5969.6154	5103.2079	4242.2662	3388.5358	2543.8247	1710.0382	889.2289
4 δ	.0010	.0010	.0010	.0011	.0011	.0011	.0011	.0011	.0011	.0012	.0012	.0012	.0012
5 BO	9275.4724	8455.8350	7632.7818	6808.1241	5983.6645	5161.2092	4342.5836	3529.6527	2724.3483	1928.7053	1144.9140	375.3959	
5 AD	6.6875	6.7300	6.7608	6.7789	6.7827	6.7710	6.7421	6.6941	6.6249	6.5318	6.4118	6.2605	
5 NA	1.8351	1.8639	1.8937	1.9244	1.9559	1.9880	2.0205	2.0532	2.0856	2.1173	2.1476	2.1754	
5 α^2	-.7390	-.7410	-.7421	-.7421	-.7410	-.7385	-.7347	-.7294	-.7223	-.7134	-.7022	-.6884	
5 α^3	-.0316	-.0267	-.0220	-.0173	-.0128	-.0084	-.0041	.0001	.0041	.0079	.0115	.0150	
5 α^4	-.0003	-.0003	-.0003	-.0002	-.0002	-.0002	-.0001	-.0001	-.0001	.0000	.0000	.0000	
5 Total	9283.2241	8463.6608	7640.6720	6816.0677	5991.6492	5169.2212	4350.6073	3537.6706	2732.3404	1936.6489	1152.7828	383.1585	
5 δ	.0011	.0011	.0011	.0011	.0011	.0012	.0012	.0012	.0012	.0012	.0012	.0012	

Components of dissociation energy of D₂, cont.

$v \setminus J$	26	27	28	29	30	31	32	33	34	35
6 BO	7491.3056	6716.9915	5940.8959	5164.9313	4391.0320	3621.1748	2857.4066	2101.8820	1356.9159	625.0621
6 AD	6.4488	6.4556	6.4489	6.4272	6.3889	6.3321	6.2548	6.1545	6.0280	5.8713
6 NA	1.9943	2.0182	2.0425	2.0672	2.0918	2.1161	2.1396	2.1617	2.1816	2.1982
6 α^2	-7149	-7147	-7133	-7106	-7066	-7010	-6937	-6844	-6729	-6587
6 α^3	-.0247	-.0202	-.0158	-.0115	-.0073	-.0032	.0007	.0045	.0081	.0115
6 α^4	-.0003	-.0003	-.0002	-.0002	-.0001	-.0001	-.0001	-.0001	.0000	.0000
6 Total	7499.0087	6724.7301	5948.6581	5172.7034	4398.7986	3628.9187	2865.1080	2109.5182	1364.4607	632.4843
6 δ	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0012	.0013
7 BO	5832.0771	5105.1579	4378.4407	3654.0078	2934.0147	2220.7279	1516.5791	824.2437	146.7656	
7 AD	6.1200	6.0889	6.0419	5.9772	5.8929	5.7864	5.6548	5.4941	5.2990	
7 NA	2.1265	2.1437	2.1604	2.1763	2.1908	2.2033	2.2128	2.2179	2.2168	
7 α^2	-.6852	-.6823	-.6781	-.6722	-.6647	-.6551	-.6433	-.6288	-.6110	
7 α^3	-.0184	-.0142	-.0102	-.0062	-.0024	.0013	.0048	.0082	.0113	
7 α^4	-.0002	-.0002	-.0002	-.0001	-.0001	-.0001	.0000	.0000	.0000	
7 Total	5839.6198	5112.6937	4385.9547	3661.4828	2941.4313	2228.0638	1523.8081	831.3351	153.6817	
7 δ	.0012	.0012	.0012	.0012	.0012	.0013	.0013	.0013	.0013	
8 BO	4304.2739	3627.6750	2953.7705	2284.9316	1623.6985	972.8597	335.5769			
8 AD	5.6989	5.6272	5.5367	5.4251	5.2895	5.1262	4.9299			
8 NA	2.2247	2.2322	2.2377	2.2404	2.2393	2.2329	2.2191			
8 α^2	-.6490	-.6429	-.6351	-.6253	-.6132	-.5983	-.5802			
8 α^3	-.0127	-.0089	-.0051	-.0016	.0019	.0051	.0082			
8 α^4	-.0002	-.0002	-.0001	-.0001	-.0001	.0000	.0000			
8 Total	4311.5357	3634.8825	2960.9045	2291.9703	1630.6160	979.6255	342.1539			
8 δ	.0013	.0013	.0013	.0013	.0013	.0013	.0012			
9 BO	2916.6835	2294.5614	1678.4204	1071.1537	476.0326					
9 AD	5.1840	5.0683	4.9298	4.7649	4.5687					
9 NA	2.2779	2.2709	2.2591	2.2411	2.2145					
9 α^2	-.6048	-.5947	-.5823	-.5672	-.5488					
9 α^3	-.0076	-.0041	-.0008	.0024	.0054					
9 α^4	-.0001	-.0001	-.0001	-.0001	.0000					
9 Total	2923.5329	2301.3016	1685.0260	1077.5947	482.2723					
9 δ	.0013	.0013	.0013	.0013	.0012					
10 BO	1681.3804	1119.8248	568.9203	32.6359						
10 AD	4.5730	4.4076	4.2125	3.9805						
10 NA	2.2696	2.2399	2.2003	2.1471						
10 α^2	-.5506	-.5352	-.5164	-.4935						
10 α^3	-.0032	-.0001	.0028	.0055						
10 α^4	-.0001	-.0001	.0000	.0000						
10 Total	1687.6692	1125.9370	574.8194	38.2756						
10 δ	.0013	.0013	.0012	.0012						
11 BO	615.6131	124.2061								
11 AD	3.8587	3.6302								
11 NA	2.1733	2.1054								
11 α^2	-.4829	-.4596								
11 α^3	.0006	.0032								
11 α^4	-.0001	.0000								
11 Total	621.1627	129.4853								
11 δ	.0012	.0012								

Components of dissociation energy of D₂, concl.

$v \setminus J$		39	40	41	42	43
0	BO	6279.8927	5250.7613	4231.2041	3222.4282	2225.6554
0	AD	8.5281	8.5934	8.6425	8.6739	8.6862
0	NA	1.4584	1.5258	1.5951	1.6663	1.7393
0	α^2	-.8762	-.8762	-.8750	-.8727	-.8691
0	α^3	.0015	.0064	.0111	.0156	.0200
0	α^4	-.0001	-.0001	.0000	.0000	.0000
0	Total	6289.0044	5260.0106	4240.5776	3231.9113	2235.2318
0	δ	.0009	.0010	.0010	.0010	.0011
1	BO	4533.9644	3559.1600	2595.3839	1644.0407	706.6032
1	AD	8.2416	8.2587	8.2562	8.2318	8.1833
1	NA	1.6703	1.7340	1.7993	1.8662	1.9346
1	α^2	-.8478	-.8451	-.8411	-.8356	-.8284
1	α^3	.0068	.0113	.0156	.0198	.0237
1	α^4	-.0001	.0000	.0000	.0000	.0001
1	Total	4543.0351	3568.3189	2604.6139	1653.3230	715.9165
1	δ	.0010	.0011	.0011	.0011	.0011
2	BO	2911.9039	1994.3210	1089.8176		
2	AD	7.8442	7.8063	7.7438		
2	NA	1.8613	1.9199	1.9794		
2	α^2	-.8132	-.8072	-.7995		
2	α^3	.0114	.0156	.0195		
2	α^4	.0000	.0000	.0001		
2	Total	2920.8076	2003.2555	1098.7610		
2	δ	.0011	.0011	.0012		
3	BO	1421.8555	565.8621			
3	AD	7.3239	7.2217			
3	NA	2.0253	2.0757			
3	α^2	-.7708	-.7604			
3	α^3	.0154	.0191			
3	α^4	.0000	.0001			
3	Total	1430.4494	574.4182			
3	δ	.0012	.0012			